



**Learning
Brains**

TRAINING ITINERARY

Learning Brains – Integrated On the-job
Learning Systems for Industrial Reskilling

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Part A: Understanding the Training Itinerary

I. Introduction to the Training Itinerary

I.1 Purpose and rationale

The Learning Brains Training Itinerary provides the curricular and pedagogical framework for the Learning Brains Training Programme, a key output of the Erasmus+ project **Learning Brains: Integrated On-the-job Learning Systems for Industrial Reskilling** (KA220-VET). The itinerary defines the structure, progression and competence development pathway through which professionals in the industrial sector can learn to use artificial intelligence (AI) to enhance on-the-job training, upskilling and reskilling processes.

This itinerary responds to a clear and growing need amongst European industrial companies, stemming from the acceleration of digital and ecological transformation processes, which is placing increasing pressure on them to upskill their staff quickly and effectively. Recent from CEDEFOP show how 45% of European workers believe that their skills could be better developed or utilised at work, 51% express a need for continuous training, and 70% of companies report a skills shortage. Considering that, workplace learning plays a key role: 69% of European workers state that they improve their skills mainly through interaction with their colleagues, and 34% indicate that they have experienced significant skills development since starting work.

In this context, AI has become a facilitator of learning in the workplace. AI-based tools make easier the creation of personalised learning modules, provide immediate learning responses accessible via mobile devices when they are needed, help to generate more engaging and interactive content, adapt to learners' individual needs, offer immediate feedback and assessment, and support the strategic planning of skills development and reskilling processes. However, many industrial companies, particularly SMEs, lack the knowledge, confidence and practical guidance needed to make the most of these tools and integrate them effectively into their training processes.

The Learning Brains Training Itinerary addresses this gap by providing a structured, progressive and practical learning pathway. It is designed not as a theoretical overview of AI, but as an applied framework that helps participants understand where AI can add value in workplace learning, how it can be used responsibly and effectively, and how it can be integrated into everyday company training practices.

1.2 The four fundamental questions

The itinerary is organised around four fundamental questions that guide its design and implementation:

1. What should professionals in the industrial sector learn about workplace training and AI?

The itinerary identifies the knowledge, skills and attitudes that HR managers, training managers, company trainers and other professionals need to develop in order to use AI effectively in on-the-job training contexts.

2. Why is each proposed topic relevant to workplace training?

Each unit in the itinerary is explicitly linked to practical workplace learning challenges and opportunities in industrial companies, ensuring that the content remains relevant, applicable and grounded in real organisational needs.

3. In what order should the areas of knowledge and learning topics be presented?

The itinerary follows a logical progression from awareness and understanding to application and improvement, ensuring that learners build competence gradually and coherently across six interconnected units.

4. How does the training pathway help learners apply AI in practice?

The itinerary is designed to support transfer into practice. It emphasises practical application, contextual adaptation, ethical and responsible use, and continuous improvement, enabling learners to move from understanding AI to using it meaningfully in their own training contexts.

By addressing these questions, the itinerary becomes the focal point of the Learning Brains Training Programme and a shared reference point for the entire partnership, as well as for external trainers and organisations interested in adapting the approach to their own contexts.

1.3 How to use this document

The itinerary serves two complementary purposes. For the Learning Brains **consortium**, it provides a shared curricular framework that will guide the development of the full Training Programme and Toolkit in subsequent project stages. For **external trainers and organisations**, it offers a practical reference that can be adapted to the design of context-specific training pathways on AI-supported workplace learning.

Thus, readers may approach this document in different ways depending on their needs:

- **Consortium partners** developing training content should read the full document, paying particular attention to the unit templates (Part B) and competence framework (Section 5), which will support content creation.
- **Trainers and training providers** interested in adapting the itinerary may focus on this Part A for understanding the overall approach, and Part C for practical guidance on adaptation, learner profiles and delivery formats.
- **HR and training managers** seeking to understand what the itinerary offers may start with Sections 1–4, which explain the purpose, target groups, learning approach and unit overview.

The document is structured to allow both linear reading and selective navigation. While Parts A and B are more descriptive and foundational, Parts C and D are more practical and orientated towards application and adaptation.

I.4 How this document fits into the Learning Brains project

The Learning Brains project brings together six partner organisations from Spain, Italy, Portugal, Austria and Slovakia to support industrial companies in the effective use of AI for on-the-job training.

The project is structured around four work packages:

- **WP1 - Project Management:** Coordination, monitoring, quality assurance and evaluation.
- **WP2 - Communication, dissemination and sustainability:** Visibility, outreach and long-term use of project results.
- **WP3 - Learning Brains Training Programme on the application of AI tools for on-the-job training:** Development of the training itinerary, the modular online course and related learning resources.
- **WP4 - Learning Brains Toolkit for the implementation of on-the-job training applying AI-based tools:** Creation of a practical, self-guided resource with step-by-step guidance, case studies, templates and ready-to-use materials for HR and training managers.

This itinerary document is a WP3 core output. It provides the curricular backbone for the Training Programme and supports the design of the Toolkit in WP4. Together, these resources enable companies and trainers to move from understanding AI to implementing it in training processes in a structured, responsible and effective way.

2. Target Groups

The Learning Brains Training Itinerary is primarily designed for professionals who play a role in planning, organising, delivering or supporting learning processes within industrial companies. The main target groups are:

- **HR managers:** Responsible for workforce development, skills planning and alignment of training with organisational strategy.
- **Training managers and training leaders:** Responsible for designing, coordinating and evaluating training programmes and learning initiatives.
- **Intermediate managers involved in training and staff development:** Team leaders, supervisors and middle managers who support learning and performance improvement in their areas.
- **Company trainers and in-company learning facilitators:** Professionals who design and deliver training activities within the company.
- **Other professionals supporting learning and reskilling processes in industrial SMEs:** Including quality managers, health and safety trainers, onboarding coordinators and others involved in workplace learning.

The itinerary is designed to be accessible to participants with varying levels of experience of using digital tools and AI. To this end, it comprises three sections: an introductory one, a development one and an application-oriented section. This means that both beginners and more experienced users can benefit from it.

Beyond the primary audience, the itinerary is also intended to be used by:

- **VET trainers and training providers** who may adapt the itinerary to design tailored programmes for companies or sectors.

- **Industry associations and clusters** who may use the itinerary to support member companies in digital transformation and skills development.
- **Policy makers and VET system actors** who may draw on the itinerary to inform initiatives on AI, digital skills and workplace learning.

3. Learning Approach

The Learning Brains Training Itinerary follows a progressive learning approach that moves from awareness to application and improvement. Participants are first introduced to the role of AI in workplace learning, then supported in identifying appropriate tools and use cases, before moving towards the design, implementation and evaluation of AI-supported training practices in company settings.

This progression reflects the project's rationale that AI should be approached as a practical support for workplace training rather than as an abstract or purely technological topic. The itinerary therefore combines conceptual understanding, practical exploration, responsible use and reflective improvement, enabling learners to connect new knowledge with concrete organisational challenges and opportunities.

The itinerary is structured into six interrelated units, each addressing different aspects of AI-supported on-the-job training. Each unit has a clear objective that distinguishes it from the others, while building on the previous units and preparing the ground for the following ones, thereby creating a coherent and cumulative learning pathway:

1. **Understanding the role of AI in on-the-job training:** Strategic introduction to AI concepts, business value and potential impact on workforce development in industrial companies.
2. **Identify relevant AI tools and use cases:** Practical exploration of AI tool categories, functions and selection criteria for different training needs and contexts.
3. **Designing AI-supported training activities:** Pedagogical application of AI tools to create learner-centred, goal-oriented training experiences.
4. **Implementing AI in company learning processes:** Organisational integration of AI tools into existing training workflows, systems and routines.

5. **Applying ethical, legal and inclusive principles:** Critical consideration of responsible AI use, including data protection, transparency, fairness, accessibility and inclusion.
6. **Evaluating results and improving practice:** Systematic assessment of training effectiveness and continuous improvement based on evidence and feedback.

This sequence reflects the project's aim of helping companies in the industrial sector to use AI as a practical tool for training, upskilling and improving performance, rather than as an abstract technology. The programme can be delivered in various formats, such as face-to-face workshops, blended learning programmes, online courses, and even self-directed learning supported by a facilitator. It is also designed to adapt to different learner profiles, organisational contexts and sector-specific realities, whilst maintaining the overall progression and the skills-based logic of the framework.

4. Overview of the Learning Units

The six learning units are described below, with their core focus and main learning outcome:

Each learning unit has a clear focus and a specific contribution to the overall curriculum logic. Together, the six units provide a balanced framework through which learners can understand the potential of AI, select relevant tools, design suitable activities, embed AI into company learning processes, apply ethical and inclusive principles, and assess results in order to improve practice over time.

Unit	Title	Core focus	Main learning outcome
1	Understanding the role of AI in on-the-job training	Introduction to the potential, limits and strategic value of AI in workplace learning, with particular attention to industrial training contexts.	Learners understand how AI can support on-the-job training and how it relates to workforce development in industrial companies.
2	Applying ethical, legal and inclusive principles	Consideration of responsible AI use, including data protection, transparency, fairness, accessibility and inclusion.	Learners can apply ethical, legal and inclusive principles when selecting, designing and using AI-supported training solutions.

3	Identify relevant AI tools and use cases	Exploration of current AI tool categories, functions and selection criteria relevant to workplace learning and training support.	Learners can identify relevant AI tools and match them to concrete training needs and company contexts.
4	Designing AI-supported training activities	Design of learner-centred, goal-oriented training activities that integrate AI tools in pedagogically meaningful ways.	Learners can design AI-supported training activities that respond to workplace learning objectives and employee needs.
5	Implementing AI in company learning processes	Integration of AI tools into company learning workflows, training routines and organisational systems.	Learners can plan the practical implementation of AI in company learning processes in a structured and context-sensitive way.
6	Evaluating results and improving practice	Evaluation of training effectiveness, collection of feedback and evidence-based improvement of future practice.	Learners can evaluate AI-supported training actions and use the results to refine and improve implementation over time.

5. Learning Brains Competence Framework

The **Learning Brains Competency Framework** defines the key **knowledge, skills and attitudes** to be developed across the six learning units. It provides a common reference point for curriculum design and ensures consistency between the training pathway, the learning objectives of each unit and the practical application of the programme in the workplace.

The framework adopts a knowledge, skills and attitudes (KSA) structure to describe what participants are expected to understand, what they should be able to do, and what professional attitudes should guide their practice. This structure contributes to the project's objective of improving digital and pedagogical capacity, whilst promoting critical, responsible and inclusive approaches to the use of AI in corporate training.

The competence framework is cumulative, and knowledge, skills and attitudes are developed progressively across the six interconnected units.

5.1 KSA matrix across the six learning units

Unit	Knowledge (K) — What learners understand	Skills (S) — What learners are able to do	Attitudes (A) — What learners value and demonstrate
1. Understanding the role of AI in on-the-job training	• Basic concepts of AI and their relevance to workplace learning. • The role of AI in supporting upskilling, reskilling and on-the-job training. • The opportunities and limitations of AI in industrial learning contexts.	• Explain the role of AI in workplace learning. • Distinguish between general AI discourse and practical training applications. • Identify areas in which AI may add value to company learning processes.	• Openness to innovation. • Willingness to explore new approaches to workplace learning. • Critical awareness of both potential and limitations.
2. Applying ethical, legal and inclusive principles	• Ethical issues linked to AI in training. • Legal and data protection considerations. • Principles of transparency, fairness, accessibility and inclusion in digital learning environments.	• Assess the ethical and legal suitability of AI-supported training practices. • Identify risks. • Apply inclusive and responsible criteria when selecting and using AI tools.	• Ethical responsibility. • Respect for learners' rights and diversity. • Commitment to fairness, inclusion and responsible innovation.
3. Identify relevant AI tools and use cases	• Different categories of AI tools relevant to training. • Typical use cases in industrial learning environments. • The relationship between training needs, user profiles and tool selection.	• Identify relevant AI tools. • Compare tools according to purpose and context. • Match specific tools to concrete learning and training needs.	• Curiosity. • Practical orientation. • Readiness to explore solutions based on relevance rather than novelty alone.

4. Designing AI-supported training activities	• Principles of learner-centred workplace training design. • Ways in which AI can support content creation, adaptation, feedback and personalisation. • Factors affecting feasibility and quality in training design.	• Design AI-supported training activities. • Formulate learning objectives. • Select suitable methods, tools and formats. • Adapt training activities to employee needs and workplace realities.	• Creativity. • Learner-centred thinking. • Openness to experimenting with AI tools. • Critical evaluation of the suitability of AI-supported design choices.
5. Implementing AI in company learning processes	• Organisational conditions for implementation. • Common barriers and enabling factors. • The relationship between tools, workflows, staff roles and company learning systems.	• Plan the introduction of AI into learning processes. • Organise implementation steps. • Anticipate barriers. • Support integration into existing training practices and organisational routines.	• Responsibility. • Solution-oriented thinking. • Willingness to manage change in a realistic and collaborative manner.
6. Evaluating results and improving practice	• Basic principles of evaluation in workplace learning. • The role of feedback, indicators and evidence in judging effectiveness. • Continuous improvement approaches in training practice.	• Evaluate AI-supported training actions. • Collect and interpret feedback. • Identify strengths, limitations and areas for improvement. • Refine future practice based on evidence.	• Reflectiveness. • Commitment to quality improvement. • Willingness to learn from experience and adapt practice over time.

Part B: Learning Units Template

6. Learning Units Template

6.1 Introduction

The Learning Brains training itinerary is intended as a reference for the further development of the Learning Brains Training Programme. At this stage, the aim is not to draw up the full training content for each unit, but rather to establish a common structure that will help all participating organisations to develop content that is coherent, comparable and aligned from a pedagogical perspective.

Therefore, each learning unit must be described using a common template. This template clarifies the role of each unit within the pathway, the expected learning outcomes, the competences to be developed, the key concepts to be addressed and guidelines for the future development of content. In this regard, the template fulfils two complementary functions: firstly, it helps the consortium to maintain consistency when designing the complete training materials in the next phase of the project; and secondly, it ensures that the pathway is understandable and transferable to other trainers or vocational training organisations wishing to adapt the Learning Brains approach to their own training contexts.

The template should accordingly be seen as a framework for the structured development of the units, rather than as a definitive teaching script. It specifies what each unit should cover and its place within the overall programme, while allowing sufficient flexibility for the subsequent creation of learning activities, case studies, examples, multimedia materials and supporting resources.

The proposed common structure for all learning units is as follows:

1. Unit overview
2. Learning outcomes
3. Competence focus
4. Key concepts and background
5. Guidance for later content development
6. Suggested methodological orientation
7. Reflection and self-assessment

Unless otherwise specified, all units address the common target groups defined earlier in this itinerary.

6.2 Example of application: Unit I

Unit 1. Understanding the role of AI in on-the-job training

1. Unit overview

This unit opens the Learning Brains itinerary by establishing a shared understanding of what artificial intelligence means in the context of workplace learning and why it is becoming increasingly relevant for industrial companies. It serves as the entry point to the whole pathway, helping participants move from a general awareness of AI towards an initial practical understanding of its relevance for on-the-job training, upskilling and reskilling processes.

At this stage, the unit is not intended to provide advanced technical knowledge. Its main role is to build a common conceptual foundation that will enable participants to engage more confidently with the following units, which focus on tools, design, implementation, ethics and evaluation.

2. Learning outcomes

By the end of this unit, participants should be able to:

- Understand the basic meaning of AI in relation to workplace learning.
- Recognise why AI is relevant to on-the-job training in industrial companies.
- Identify key opportunities and challenges linked to the use of AI in training processes.
- Distinguish between general discourse about AI and its practical application in company learning contexts.
- Begin to reflect on how AI may support training, reskilling and continuous development in their own organisational context.

3. Competence focus

Knowledge

- Basic concepts related to AI and AI-supported learning.
- The role of AI in workplace learning and continuous vocational training.
- Main opportunities, benefits and limitations of AI in industrial training contexts.
- The relationship between AI, digital transformation and workforce development.

Skills

- Explain in simple terms how AI can support on-the-job training.
- Recognise potential areas where AI may add value in company learning processes.
- Relate broad AI concepts to practical workplace learning challenges.
- Frame initial discussions about AI in a realistic and professionally relevant way.

Attitudes

- Openness to innovation in training practice.
- Curiosity about the potential of AI in workplace learning.
- Critical awareness of both opportunities and risks.
- Willingness to explore AI as a support tool rather than an end in itself.

4. Key concepts and background

This unit should introduce the basic ideas that participants need in order to understand the rest of the itinerary. These include the general meaning of artificial intelligence, the idea of AI as a support for learning and training processes, and the reasons why AI is becoming increasingly relevant in industrial and SME contexts.

The background to the unit should also reflect the broader project rationale. Learning Brains is based on the recognition that industrial companies operate in conditions of digital and green transition, rapid technological change and increasing skills needs, and that workplace learning plays a central role in helping employees adapt to these changes. In this context, AI is presented not as a purely technical innovation, but as a practical resource that may help companies create more responsive, personalised and efficient training opportunities.

5. Guidance for later content development

When the consortium develops the full Training Programme, this unit may later include:

- Introductory explanations of AI terminology.
- Simple examples of AI use in workplace learning.
- Short industrial scenarios showing possible applications.
- Short video or audiovisual explainers.
- Reflection prompts linked to participants' own company reality.

- Introductory case examples that prepare participants for later units on tools and implementation.

At itinerary stage, these elements should be understood as indicative directions for content creation rather than as finalised materials.

6. Suggested methodological orientation

This unit is especially suitable for:

- Introductory self-paced learning.
- Guided reflection on participants' own training reality.
- Short scenario-based examples.
- Myth-versus-reality discussions about AI.
- Activities that connect broad concepts with concrete workplace situations.

The methodological approach should focus on accessibility and building confidence. As an introductory unit, its purpose is to help participants develop a shared vocabulary and practical frame of reference, while avoiding the inclusion of excessive technical detail.

7. Reflection and self-assessment

Indicative reflection prompts may include:

- What training challenges currently exist in the participant's organisation?
- In what ways might AI support learning or training processes in that context?
- What opportunities seem most relevant or promising?
- What doubts, barriers or risks would need to be considered?

Indicative self-assessment statements may include:

- I understand the basic meaning of AI in the context of workplace learning.
- I can explain why AI is relevant to on-the-job training.
- I can identify initial opportunities and challenges linked to AI-supported training.
- I can relate the topic of AI to the training needs of my organisation.

Part C: Practical Guidance for Trainers

7. How to Use and Adapt the Itinerary

The Learning Brains Training Itinerary is designed to fulfil a double function within the project. On the one hand, it provides the consortium with a shared curricular framework that supports the coherent development of the Learning Brains Training Programme. On the other hand, it is also intended to serve as a practical reference document for external VET trainers, training providers, company trainers and organisations interested in designing their own AI-related training pathways for on-the-job learning in industrial contexts.

For this reason, the itinerary should not be read as a fixed course that must be reproduced in exactly the same way in every setting. Instead, it should be understood as a structured framework that defines the main learning logic, the progression of topics, the competence areas to be developed and the general principles that should guide the later creation of training contents, activities and supporting materials.

Users of the itinerary are encouraged to adapt it to their own organisational, sectoral and learner realities. Such adaptation may involve selecting only certain units, adjusting the depth of treatment of each topic, integrating local examples or sector-specific cases, or choosing different delivery modes according to time, resources and participants' needs. What should remain constant, however, is the overall progression of the itinerary, which moves from understanding AI and identifying its practical potential to designing, implementing, assessing and improving AI-supported workplace learning practices.

When using the itinerary to create a training programme, trainers and organisations should begin by identifying the specific professional profile of their learners, the main training needs to be addressed and the level of prior familiarity with AI and digital tools. On this basis, they can decide whether to implement the full itinerary, to prioritise selected units, or to combine the itinerary with existing resources, examples and materials available in their own context.

The itinerary is therefore both a roadmap and an adaptation tool. Its value lies not only in what it proposes, but also in the way it helps users create coherent, context-sensitive and practically relevant learning pathways based on a common pedagogical structure.

8. Adapting to Different Learner Profiles

One of the strengths of the Learning Brains itinerary is that it can be used with different professional profiles involved in workplace learning and staff development. However, the emphasis placed on each unit may vary depending on the learners' role, responsibilities and degree of influence over training processes within their organisation.

8.1 Examples of learner profiles in the industrial sector

Different types of learners may require different entry points and different emphases within the itinerary. The following examples illustrate how this may work in practice.

HR managers and training managers

These profiles are often responsible for planning staff development, identifying skills gaps, selecting training solutions and aligning learning actions with wider business or workforce strategies. For them, the itinerary should place stronger emphasis on:

- Identifying use cases for AI in training.
- Selecting relevant tools.
- Planning implementation.
- Ensuring ethical and legal compliance.
- Evaluating organisational impact.

For example, an HR manager may need to understand how AI could support onboarding, internal mobility or reskilling pathways across departments, but may not need highly detailed guidance on facilitating every training activity. In this case, Units 2, 4, 5 and 6 may be given greater weight than highly operational training design details.

Company trainers and in-company learning facilitators

These learners are closer to the practical delivery of training and are more likely to need support with designing activities, creating or adapting content, using tools in a pedagogically meaningful way, and responding to different learner needs. For them, the itinerary should place stronger emphasis on:

- Practical tool use.
- Activity design.
- Learner support.
- Accessibility and inclusion.
- Formative feedback processes.

For example, a company trainer may need to learn how to use AI to generate differentiated learning materials, create short just-in-time support resources for workers on the shop floor, or adapt explanations to workers with different qualification levels. In this case, Units 2, 3, 5 and 6 may be particularly central.

Middle managers, supervisors and team leaders

These profiles are often involved in staff development, but not necessarily as formal trainers. Their main concern is usually operational: how to help workers perform better, adapt to change and solve problems in real work situations. For them, the itinerary may need to be shorter, more applied and strongly linked to immediate workplace challenges.

For example, a production supervisor may not need a broad overview of AI in training systems but may benefit from practical exposure to simple uses such as creating step-by-step guidance, supporting onboarding of new workers, or identifying which training issues could be addressed through short AI-supported learning interventions. In these cases, the programme may focus on Units 1, 2 and 4, with selected elements from Units 5 and 6.

Skilled workers and technicians

Where the itinerary is adapted for skilled workers, the emphasis may shift from training design to practical use of AI-supported learning resources. These learners may not be expected to design a training programme themselves, but they may need to understand how to access, interpret and use AI-enabled learning support in their daily tasks.

For example, maintenance technicians, machine operators or quality-control staff may benefit from training on how AI-supported resources can help them troubleshoot, refresh procedures, access personalised learning support or learn updated processes more quickly. In this case, the content should be concrete, task-oriented and clearly connected to real work situations rather than presented as a general course on AI.

Lower-qualified workers or workers with lower digital confidence

Some learner groups may need a much more gradual, confidence-building approach. This may apply to workers with fewer formal qualifications, workers with limited previous use of digital tools, or workers who feel unsure about AI and its implications. For these learners, trainers may need to reduce jargon, use clearer examples, slow the pace, provide more guided support and focus on practical benefits that are easy to recognise.

For example, instead of beginning with categories of AI tools, a trainer might begin with familiar workplace problems such as "How can a worker quickly access the right instruction at the right time?" or "How can learning materials be explained more simply?" and only then introduce AI as one possible support. In such cases, Units 1 and 2 may need more time, and Unit 3 may need to be adapted in a more guided and simplified form.

Older workers or learners more reluctant to engage in training

Age itself should not be treated as a deficit factor, but some older workers or long-serving staff may be more sceptical about new training approaches, less confident with digital systems, or less convinced of the value of further training. In these cases, resistance may be linked less to age itself than to previous training experiences, fear of replacement, lack of time or uncertainty about relevance.

For these learners, adaptation should focus on:

- Building trust.
- Demonstrating immediate usefulness.
- Avoiding overly technical language.
- Acknowledging existing expertise.
- Showing how AI can support rather than replace professional knowledge.

For example, a trainer might begin from the message that AI can help capture and transfer experienced workers' knowledge, support mentoring or make procedures easier to update, rather than framing AI as a disruptive change imposed from outside. This can reduce defensiveness and increase engagement.

8.2 Factors to consider when adapting the itinerary

When adapting the itinerary, trainers or programme designers should consider at least the following variables:

- **Professional role:** Is the learner expected to plan training, deliver it, support it, or mainly benefit from it as an end user?
- **Work area:** Is the training aimed at HR, production, maintenance, quality, logistics, onboarding, health and safety, or another area of company activity?
- **Qualification level:** Are learners highly qualified specialists, operational staff, or workers with fewer formal qualifications?
- **Digital competence:** Are learners already comfortable with digital tools, or do they need a gradual introduction?
- **Attitude towards training and innovation:** Are learners motivated and curious, or sceptical, reluctant or anxious about change?
- **Language and literacy demands:** Does the programme need simpler language, more visual support, or more guided facilitation?
- **Time and organisational conditions:** Are learners available for a full programme, or only for short modular interventions linked to work routines?

8.3 A basic adaptation roadmap

In order to make these adaptations more systematic, the person designing the programme may use a simple sequence of questions before selecting units, methods or materials.

Step 1. Who are the learners?

- What is their professional role in the company?
- What is their involvement in training processes? Are they responsible for planning training, delivering it, supporting it, or mainly benefiting from it as participants?
- What is their expected role in applying AI-supported training after the programme? Will they design training activities, facilitate learning processes, or use AI-supported learning resources in their daily work?

Step 2. What do they actually need?

- What workplace learning problem or opportunity is the programme intended to address?
- Is the primary need strategic (e.g., workforce development planning), pedagogical (e.g., designing better training), operational (e.g., improving onboarding), or task-specific (e.g., just-in-time support for workers)?
- Which units of the itinerary are most relevant to that need?

Step 3. What is their starting point?

- What is their current level of digital competence and confidence with digital tools?
- How familiar are they with AI?
- How confident are they in engaging with new tools or new learning formats? Are they likely to approach the topic with curiosity, scepticism or anxiety?

Step 4. What may support or hinder engagement?

- What barriers may exist, such as lack of time, fear of complexity, low confidence or unclear relevance to their daily work?
- What organisational conditions may affect implementation, such as available resources, management support or existing training infrastructure?
- What examples, entry points or framing could make the training feel immediately useful and credible to this group?

Step 5. How should the programme be adapted?

- Which units should be prioritised, and which may be condensed or omitted based on learner needs and available time?
- Which concepts need simplification, expansion or stronger practical examples for this specific group?
- What language, pacing, delivery format and methodology will best suit their profile and context?

Step 6. What evidence will show that the adaptation was appropriate?

- How will the trainer know whether the level, format and examples were suitable for this group?
- What feedback should be collected from learners during and after the programme?
- What signs of engagement, application or improvement should be observed to judge the success of the adaptation?

9. Pedagogical Recommendations for Trainers

The Learning Brains itinerary is intended for professional and vocational learning contexts in which participants usually expect practical value, relevance to real work situations and direct applicability. For this reason, training based on the itinerary should avoid excessively abstract or overly technical approaches and should instead connect AI-related concepts with concrete workplace challenges, decisions and opportunities.

1. Start from participants' realities

A first recommendation is to start from participants' realities rather than from technology itself. Trainers should encourage learners to reflect on their training needs, operational problems or skills gaps present in their own company or sector and then explore how AI may or may not help address them. This helps to frame AI as a support for workplace learning and performance improvement.

2. Balance exploration with critical reflection

A second recommendation is to balance exploration with critical reflection. While the itinerary aims to promote the adoption of AI-supported approaches in training, it also recognises the importance of responsible use, realistic expectations and attention to ethical, legal and inclusive considerations. Trainers should therefore create space not only for enthusiasm and experimentation, but also for discussion of limitations, risks, barriers and conditions for appropriate use.

3. Privilege examples, cases and scenarios

The third recommendation is to prioritise examples, cases and scenarios over generic explanations wherever possible. As many learners will not be technical experts, they are more likely to understand the practical value of AI through realistic situations, decision-making scenarios, workplace examples and simple demonstrations than through abstract definitions alone. This is particularly important in industrial settings, where relevance to processes, tasks and organisational realities is vital.

4. Use the itinerary as a guide

Finally, trainers should use the itinerary as a guide rather than a script. The purpose of the document is to guide the design of coherent learning pathways, not prescribe a single teaching method. Trainers should therefore feel empowered to adapt methods, timings, examples and resources to suit their audience, while maintaining the progression and competence logic set out in the itinerary.

10. Delivery Formats and Methodological Options

The Learning Brains itinerary is intended to be flexible enough to support different delivery formats, depending on the time available, the profile of the participants, the learning environment and the resources of the provider or organisation. For this reason, it should be seen as compatible with face-to-face, online, blended and self-directed formats, provided that the coherence of the learning pathway is maintained.

1. Face-to-face formats

In face-to-face formats, the itinerary may be implemented through workshops, short training cycles or facilitated group sessions. This format is especially well suited to discussion, collaborative reflection, peer exchange, joint problem-solving and the use of practical examples drawn from participants' organisations. It may be particularly useful for introductory units, ethical discussion and activities that require exchange of perspectives.

2. Blended learning formats

In blended learning formats, trainers may combine self-paced input with synchronous workshops or coaching sessions. For example, participants may first explore concepts, examples or short digital resources individually, and then meet in person or online to discuss use cases, compare tools, apply ideas to their context or reflect on implementation challenges. This approach can support flexibility while still preserving interaction and guided reflection.

3. Online or self-directed formats

In online or self-directed formats, the itinerary may serve as a modular pathway in which each unit is translated into shorter learning blocks, guiding questions, practical tasks and self-assessment activities. This can be useful when learners have limited time or when training needs to be integrated into ongoing professional activity. In these cases, however, it is especially important to ensure clarity of structure, relevance of examples and opportunities for meaningful application to learners' own contexts.

4. Alternative methodological options

Alternative methodological options may also be considered according to the aim of the programme. A trainer may use scenario-based learning to connect AI to workplace decisions, case-based learning to illustrate implementation choices, peer exchange to surface practice-based insight, or reflective tasks to support strategic planning and responsible adoption. The itinerary does not impose a single methodology, but it does encourage methods that are participatory, context-aware and oriented towards practical transfer.

11. Sequencing, Pacing and Level of Depth

Although the Learning Brains itinerary is structured as a six-unit progression, this does not mean that every implementation must dedicate the same amount of time or level of detail to each unit. Different users may require different degrees of depth depending on their objectives, starting point and available time.

In some cases, the full sequence may be implemented as originally conceived, especially where the aim is to build comprehensive capability across understanding, design, implementation, ethics and evaluation. In other cases, the itinerary may be condensed into a shorter programme in which the most immediately relevant units are prioritised, provided that enough introductory grounding is retained to ensure coherence.

Trainers may also adapt pacing according to learner profile. Groups with little familiarity with AI may require more time in Units 1 and 2, while more experienced audiences may move more quickly through introductory material and spend more time on design, implementation and evaluation. Similarly, groups focused on organisational planning may give greater attention to Units 4 to 6, while trainer-centred groups may emphasise Units 2 to 5.

The key principle is that adaptation should remain intentional rather than arbitrary. Even when the duration, depth or order of emphasis is modified, the trainer should preserve the overall developmental logic of the itinerary and make explicit how the selected content contributes to the broader competence pathway.

I2. Contextualising to Sectors and Organisations

The Learning Brains itinerary is developed with a strong focus on industrial SMEs and workplace learning in contexts of digital and organisational change. Nevertheless, its structure is broad enough to be adapted to different industrial subsectors, organisational realities and training ecosystems.

In practice, contextualisation may involve selecting examples that reflect a particular industry, such as manufacturing, metal, machinery, construction-related environments or other technical and production-oriented sectors represented in the project context. It may also involve adapting the language used in the programme so that it speaks directly to the concerns of the learner group, whether this concern productivity, onboarding, quality, safety, maintenance, upskilling or change management.

Organisational context also matters. A small company with limited digital maturity may need a more gradual and confidence-building approach, with emphasis on simple use cases and immediate benefits. A more advanced organisation may be ready to engage with more complex design, implementation and evaluation issues, including strategic integration and continuous improvement. The itinerary should therefore be used as a stable structure that supports variation in examples, complexity and application according to context.

This contextual flexibility is important for transferability. By combining a common competence and progression framework with room for local adaptation, the itinerary can remain both coherent and widely usable across different organisations, countries and VET settings.

Part D: European Reference Alignment

I3. European Reference Alignment

The Learning Brains Training Itinerary can also be read in dialogue with wider European and international frameworks on digital competence, AI and training practice. While the itinerary is not intended to replicate these frameworks directly, it is compatible with their general orientation and can benefit from being positioned in relation to them at document level rather than unit by unit.

Aligning the itinerary with established European frameworks serves several purposes.

Firstly, it reinforces the credibility and transferability of the Learning Brains approach by connecting it to recognised standards. Secondly, it helps users understand how the itinerary relates to broader policy initiatives on digital skills, AI literacy and continuous vocational training. Thirdly, it provides a common language that can facilitate dialogue between different stakeholders, including companies, training providers, policy makers and researchers.

The three main frameworks referenced in this document are DigComp 3.0, DigCompEdu and the UNESCO AI Competency Framework for Teachers. Each contributes a different perspective that enriches the itinerary's conceptual foundation.

13.1 DigComp 3.0: European Digital Competence Framework

DigComp 3.0 is the most recent iteration of the European Digital Competence Framework for Citizens, published by the European Commission's Joint Research Centre. It describes the knowledge, skills and attitudes that are needed to be digitally competent for daily life, participation in society, working and learning. The framework is technology-neutral and is designed to be tailored and adapted for a variety of purposes in education, training and employment contexts.

DigComp 3.0 reflects rapid digital technological developments that have occurred since 2022, and which have wide-ranging implications for digital competence. It incorporates new learning outcomes that offer a more detailed view of digital competence than before, alongside the systematic and cross-cutting integration of AI throughout the framework. This makes DigComp 3.0 a particularly relevant reference for Learning Brains, as its itinerary also seeks to integrate AI competence across multiple dimensions of workplace learning practice.

The Learning Brains itinerary aligns with DigComp 3.0 in several key respects:

- **Information and data literacy:** The itinerary supports learners in identifying, accessing and evaluating AI tools and resources, which corresponds to DigComp's emphasis on critical engagement with digital information.
- **Communication and collaboration:** By encouraging learners to consider how AI can support workplace learning and knowledge sharing, the itinerary connects to DigComp's focus on digital communication and collaborative practices.

- **Digital content creation:** Units 3 and 4 of the itinerary, which address the design and implementation of AI-supported training activities, align with DigComp's competence area on creating and adapting digital content.
- **Safety:** Unit 2 of the itinerary, which addresses ethical, legal and inclusive principles, corresponds to DigComp's emphasis on protecting devices, content, personal data and privacy, as well as health and well-being in digital environments.
- **Problem solving:** The itinerary's overall orientation towards practical application and workplace challenges reflects DigComp's focus on identifying needs and problems, and resolving them through digital means.

By positioning the Learning Brains itinerary in relation to DigComp 3.0, the document contributes to broader European efforts to strengthen digital skills and AI literacy across education, training and employment contexts.

13.2 DigCompEdu: European Framework for the Digital Competence of Educators

DigCompEdu is the European Framework for the Digital Competence of Educators, also developed by the European Commission's Joint Research Centre. It is directed towards educators at all levels of education, from early childhood to higher and adult education, including general and vocational education and training, special needs education, and non-formal learning contexts.

DigCompEdu describes 22 educator competences organised across six areas:

5. **Professional Engagement:** Using digital technologies for professional collaboration, reflective practice and professional development.
6. **Digital Resources:** Sourcing, creating and sharing digital resources for teaching and learning.
7. **Teaching and Learning:** Managing and orchestrating the use of digital technologies in teaching and learning.
8. **Assessment:** Using digital technologies for formative and summative assessment.
9. **Empowering Learners:** Using digital technologies to support learners' inclusion, personalisation and active participation.

10. Facilitating Learners' Digital Competence: Enabling learners to creatively and critically use digital technologies.

The Learning Brains itinerary is particularly relevant to several of these areas, especially when the itinerary is used by VET trainers, company trainers or training providers designing AI-supported learning experiences:

- **Digital Resources (Area 2):** Units 3 and 4 of the itinerary address the identification of AI tools and the design of AI-supported training activities, and correspond closely to DigCompEdu's focus on selecting, creating and adapting digital resources for learning.
- **Teaching and Learning (Area 3):** Units 4 and 5 address the design and implementation of training activities, aligning with DigCompEdu's emphasis on managing and using digital technologies in teaching and learning processes.
- **Assessment (Area 4):** Unit 6 of the itinerary addresses evaluation and improvement of practice, and connects to DigCompEdu's focus on using digital technologies for assessment and feedback.
- **Empowering Learners (Area 5):** the Units 3 and 4 address learner-centred design, accessibility and inclusion, thus reflects DigCompEdu's commitment to using digital technologies to support all learners, including those with specific needs or fewer opportunities.
- **Facilitating Learners' Digital Competence (Area 6):** By supporting trainers in developing AI-supported learning experiences, the itinerary contributes to the broader goal of enabling learners to develop their own digital competence.

DigCompEdu is primarily oriented towards educators in formal education settings, but its competence areas are sufficiently broad to be relevant to workplace learning and C-VET training contexts. The Learning Brains itinerary can therefore be seen as a sector-specific adaptation that applies DigCompEdu principles to the particular challenges and opportunities of AI-supported on-the-job training in industrial companies.

13.3 UNESCO AI Competency Framework for Teachers

The **UNESCO AI Competency Framework for Teachers (AI CFT)**, published in 2024, offers a global vision that outlines the necessary skills and knowledge for teachers to effectively integrate artificial intelligence (AI) into their teaching practices. The framework is built on a human-centred approach, prioritising values such as inclusivity, human agency, and respect for diversity.

The AI CFT is organised around five key areas:

1. **Teachers' Mindset:** Developing a positive, critical and human-centred orientation towards AI in education.
2. **AI Ethics:** Understanding ethical issues, risks and responsible practices related to AI use in educational contexts.
3. **AI Foundations and Applications:** Understanding what AI is, how it works, and how it can be applied in teaching and learning.
4. **AI Pedagogy:** Using AI to support teaching, learning and assessment in pedagogically meaningful ways.
5. **AI for Professional Development:** Using AI to support teachers' own professional learning and development.

The framework also defines 15 competencies across these five areas and organises them across three progression levels: acquire, deepen and create.

The Learning Brains itinerary aligns closely with the UNESCO AI CFT in several respects:

- **Teachers' Mindset (Area 1):** Unit 1 introduces AI concepts and encourages openness balanced with critical awareness, corresponding to the AI CFT's emphasis on developing a constructive and reflective orientation towards AI.
- **AI Ethics (Area 2):** Unit 2 addresses ethical, legal and inclusive principles, aligning directly with the AI CFT's focus on ethical and responsible AI use, including attention to fairness, transparency, privacy and inclusion.
- **AI Foundations and Applications (Area 3):** Units 1 and 3 of the itinerary introduce AI concepts and explore relevant tools and use cases, corresponding to the AI CFT's focus on understanding AI foundations and applications.

- **AI Pedagogy (Area 4):** Units 4 and 5 address the design and implementation of AI-supported training activities, aligning closely with the AI CFT's emphasis on using AI in pedagogically meaningful ways to support learning.
- **AI for Professional Development (Area 5):** Unit 6 addresses evaluation and continuous improvement, connecting to the AI CFT's focus on using AI to support ongoing professional learning and development.

Although the UNESCO AI CFT is primarily oriented towards school teachers, its human-centred approach and competence structure are highly relevant to continuous vocational education and workplace learning contexts. The Learning Brains itinerary can therefore be seen as a sector-specific adaptation that applies UNESCO AI CFT principles to the particular context of AI-supported on-the-job training in industrial companies.

13.4 How the frameworks inform the itinerary

Taken together, these three frameworks help situate the Learning Brains itinerary within a wider European and international conversation on digital competence, responsible technology use and innovation in education and training.

- **DigComp 3.0** provides the broad digital competence foundation, ensuring that the itinerary contributes to general digital literacy and AI competence as understood at European level.
- **DigCompEdu** adds an educator-specific perspective, ensuring that the itinerary supports trainers and training professionals in developing the digital competences needed to design and facilitate AI-supported learning experiences.
- **UNESCO AI CFT** brings a focused AI-specific lens, ensuring that the itinerary addresses the particular knowledge, skills and attitudes needed to use AI responsibly and effectively in educational and training contexts.

Including these references at overall itinerary level helps maintain clarity while still reinforcing the credibility and transferability of the document. It also allows users to understand how the Learning Brains approach relates to wider policy and practice developments in digital skills, AI literacy and continuous vocational education.